PROJECT 3

E-Commerce Sales Analytics and Customer Churn Prediction

Course: Data Engineering and MLOps | **Mode:** Individual Project | **Total Marks:** 100

Assignment Part 1 Submission Deadline: On or before 7 September 2026 (First Week of September)

1. Understanding the Project

Build an analytics platform that consolidates online retail transactions, customer details, and product information. Part 1 focuses on a sales data warehouse and interactive business dashboard; Part 2 adds customer-churn or repeat-purchase prediction using MLOps practices.

Project objectives

- Create a reliable retail ETL pipeline.
- Design customer, product, order, and date dimensions.
- Provide sales and retention dashboards.
- Build and operationalize a churn or repeat-purchase model.

2. Data Source Information

Data source	Information available	Suggested use
UCI Online Retail Dataset	Transaction-level online retail data including invoice, product, quantity, price, customer, and country.	Primary public dataset
Kaggle e-commerce datasets	Alternative order, product, and customer datasets.	Optional source
Synthetic customer master file	Student-created customer demographic or loyalty attributes.	Optional enrichment
CSV/Excel order files	Simulated monthly extracts to demonstrate incremental ingestion.	Incremental source

Dataset rule: Use only free, legally accessible, public, institutional, or instructor-approved datasets. Include the source name and access instructions in the report.

3. Data Ingestion Tool Information

Use Python/Pandas for file ingestion and Apache Airflow for incremental processing. Students may use Pentaho Community Edition instead of Airflow if approved. Data should be loaded into PostgreSQL.

Minimum ingestion expectations

- Maintain a raw copy of the extracted data.
- Record extraction date, source, file/API status, and row count.
- Handle extraction errors and log failed records.
- Support at least one repeatable or scheduled ingestion workflow.
- Do not manually edit the final analytical dataset.

4. Suggested Data Architecture Design

```
Monthly order files/customer/product files -> landing area -> Airflow/Pentaho ETL -> staging
tables -> PostgreSQL star schema -> sales mart and customer mart -> Tableau Public dashboard.
```

Students should prepare their own architecture diagram showing the source layer, ingestion layer, raw/staging layer, transformation layer, storage layer, analytics layer, and later the MLOps layer.

5. ETL and Data Engineering Requirements

- Remove cancelled orders and invalid quantities where appropriate.
- Standardize currency and date formats.
- Create customer, product, geography, and date dimensions.
- Calculate revenue, order value, recency, frequency, and monetary features.
- Implement incremental loading and duplicate prevention.
- Create sales and retention fact tables.

6. Suggested Data Visualization

Use **Tableau Public** for business-oriented dashboards and comparative analysis, or **Streamlit** when interactive filtering, user input, or future model prediction is central to the application.

- Revenue and order trends
- Top products and categories
- Customer segmentation
- Country-wise sales
- Retention and repeat-purchase analysis

7. Assignment Part 1 - Data Pipeline Implementation

Deadline: On or before 7 September 2026, which falls in the first week of September.

Students must build a complete data pipeline without adding the final machine-learning deployment at this stage.

Required implementation

- Acquire and document the selected dataset(s).
- Create automated or reproducible ingestion scripts.
- Implement an Apache Airflow DAG, or an approved equivalent ETL workflow.
- Create raw, staging, cleaned, and analytical data layers.
- Apply data-quality checks and produce a rejected-record or error log.
- Load the transformed data into PostgreSQL or another approved open-source database.
- Create at least one analytical table or data mart.
- Develop a Tableau Public or Streamlit dashboard with at least five meaningful views or indicators.
- Prepare an architecture diagram and data dictionary.
- Demonstrate one successful end-to-end pipeline execution.

Part 1 submission package

Item	Required submission
1	Source code and ETL scripts
2	Airflow DAG or approved orchestration workflow
3	Database schema and sample populated tables
4	Dataset source information and access instructions
5	Architecture diagram and pipeline flow
6	Data dictionary and validation rules
7	Tableau Public link/file or Streamlit application
8	Project report of approximately 8-10 pages
9	Screenshots or execution evidence
10	README with setup and run instructions

8. Assignment Part 2 - MLOps Pipeline Extension

After completing Part 1, students will extend the same project into an end-to-end MLOps solution. The Part 2 deadline will be announced separately.

- Define churn using a clear inactivity window.
- Train Logistic Regression, Random Forest, or XGBoost models.
- Track experiments, parameters, and metrics with MLflow.
- Register the best model and expose a prediction API.
- Dockerize the API and dashboard.
- Monitor customer-feature drift and model precision/recall.

Common Part 2 requirements

- Create reproducible feature-engineering and model-training pipelines.

- Use a chronological or problem-appropriate train/validation/test split.
- Track experiments, parameters, metrics, and artifacts using MLflow.
- Register and version the selected model.
- Serve predictions through FastAPI or an approved API framework.
- Containerize the inference service using Docker.
- Integrate model output into the Streamlit or Tableau-supported workflow where appropriate.
- Monitor input-data quality, drift, model performance, latency, and failures.
- Define retraining criteria and document the model lifecycle.

9. Suggested Mark Distribution

Component	Part 1	Part 2
Problem understanding and design	5	5
Data source and ingestion	10	5
ETL, validation, and storage	20	5
Visualization and interpretation	10	5
Documentation and demonstration	5	5
Model development and evaluation	-	10
MLflow, registry, deployment, and Docker	-	10
Monitoring and retraining design	-	5
Total	**50**	**50**

10. General Instructions

- The project must be completed individually.
- All source code must be written, organized, and explained by the student.
- Public code may be consulted, but copied work without attribution will not be accepted.
- Use environment variables for database credentials and API keys.
- Do not upload private, personally identifiable, or confidential data.
- The final demonstration must show the pipeline from data ingestion to dashboard output.
- Late submissions will be handled according to course policy.